

Hiro Liu

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Professional Summary

- Senior Software Engineer with 13+ years of cross-platform development experience across Android, Windows, iOS, and Linux.
- Specialized in endpoint security (Trend Micro Worry-Free Business Security, 4M+ managed endpoints), industrial IoT device management, system-level debugging, and CI/CD automation.
- Proven ability to lead small teams, manage the full SDLC, and ship production-quality software in Agile/Scrum environments.
- Actively leverages AI-assisted development (Claude Code, Copilot) to accelerate delivery and enhance support workflows.

Technical Skills

- **Languages:** C++, C, Java, Python (primary) | C#, Swift, Objective-C, Node.js, Batch Script (secondary)
- **Platforms:** Windows, Android, iOS, Linux (Alpine)
- **Dev Tools:** Android Studio, Visual Studio, GitHub Actions, Docker, Appium, Android NDK/JNI
- **Diagnostic Tools:** WinDBG, Process Monitor, Process Explorer, Windows Performance Analyzer, Wireshark, VMware
- **CI/CD & DevOps:** GitHub, YAML, Python scripts, Batch Script, JFrog Artifactory
- **Expertise:** Endpoint security, cross-platform SDK/API design, device communication (UART/DIO/AIO), RESTful API, telemetry analytics (AWS Athena, Kibana)
- **Methodologies:** Agile, Scrum, DevOps, test automation, AI-assisted development

Professional Experience

Trend Micro | *Senior Software Engineer* (May 2020 – 2026)

Product: Worry-Free Business Security Services (WFBS-SVC) — SaaS endpoint protection across 80k+ Android and 4M+ Windows managed endpoints globally.

Android Platform

- Maintained and enhanced Android endpoint security agent (Java + NDK/JNI), protecting 80k+ of managed devices; core file-scanning engine implemented in native C/C++ via JNI.
- Built Android automated testing framework from scratch using Appium + Python, eliminating manual regression effort and improving release stability.
- Investigated and resolved complex customer-escalated issues using deep Android system knowledge.

Windows Platform

- Developed and maintained Windows endpoint security agent (C++) deployed across 4M+ endpoints.
- Diagnosed and resolved complex system-level issues (system hangs, performance degradation, 3rd-party conflicts) using WinDBG, Process Monitor, Process Explorer, Windows Performance Analyzer, and VMware.
- Handled **70–90 customer escalations per year**;; triaged and resolved P0/P1 critical cases requiring remote debug sessions with customers.

Cross-functional / Common

- Optimized CI/CD workflows using GitHub Actions, YAML, Python, Batch Scripts, and JFrog Artifactory.
- Leveraged telemetry data (AWS Athena SQL queries, Kibana) for large-scale endpoint monitoring and proactive issue detection.
- Followed Agile/Scrum with a consistent 1–2 release cadence per month.
- Integrated AI tools (Claude Code, Copilot) into daily development and customer-support workflows.

SUNIX Co., Ltd. | *Section Manager / Software Engineer* (Oct 2012 – Apr 2020)

Product: DevicePort / iDevicePort — industrial-grade serial/IO port virtualization & remote management solution for POS, industrial control, and IoT.

- Promoted to Section Manager after 2 years; led a 3-person team responsible for project planning, delivery scheduling, and junior engineer mentoring.
- Developed Windows management software for DevicePort LAN-based device management (C, C++, C#), applying Shared Memory, Named Pipe, multi-threading, and process management techniques.
- Designed software-hardware communication protocols and data-transfer specifications (UDP/TCP) for the iDevicePort product line.
- Built portable ****Android and iOS SDK frameworks**** (Java, Objective-C) enabling third-party integration with iDevicePort; delivered full sample applications alongside each SDK.
- Created cross-platform IoT local service packaged with Alpine Linux + Docker, exposing RESTful APIs via Node.js for flexible cloud-level device management (DeviceCloud).
- Performed network packet analysis (UDP/TCP/IP) and worked with industrial communication protocols (UART, DIO, AIO).

CyberLink Corp. | *Software Test Intern* (Jul 2010 – Sep 2010)

- Executed functional and UX test plans for multimedia software products.

Education

National Taiwan University of Science and Technology | B.S. in Computer Science and Information Engineering · 2007 – 2011

- Graduation project: Simulated FTL algorithm (BAST) for NAND Flash memory on Linux using C.

Military Service (R.O.C. Army) | Jul 2011 – Oct 2012

Core Competencies

- Strong background in ****Android development**** (application-layer agent, SDK design, automated testing framework).
- Expert-level ****system debugging and performance profiling**** across Windows and Android platforms.
- Broad cross-platform engineering background spanning IoT hardware integration and cybersecurity SaaS.
- Hands-on ****team leadership****: project planning, junior mentoring, cross-functional collaboration.
- Highly adaptable — rapid adoption of emerging technologies including AI-assisted development tooling.